CIVILIZATION — Consolidated Planning Reference

A Cybersecurity CTF tracing the root of civilization to the evolution of technology

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01 — Theme & Structure

Five eras, each mapping a stage of civilization's technological evolution onto a security concept. Subcommittee members pick their final questions from the suggestion pools in Section 02 — nothing below is a fixed roster.

Era	Civilizational Idea	Technical Theme	Target Difficulty
0. Warm-Up	Before systems existed at all	General awareness, no tools needed	Zero-barrier
I. Roots	The first knowledge systems — writing, record-keeping	Reconnaissance, hidden information, basic web	Beginner
II. Settlement	The first trust systems — trade, identity, records	Authentication, databases, access control	Easy–Medium
III. Empire	The first infrastructure at scale — administration, logistics	Privilege escalation, chained exploitation	Medium–Hard
IV. Industry & Information	Mechanization, then computing	Cryptography, client-side attacks, modern web flaws	Hard
V. Convergence	Interconnected, AI-driven systems	Multi-service pivoting, full-chain compromise	Expert (gated, unlocks mid-event)

Every era should run at least one **Lite** question alongside harder options — same concept, fewer steps, lower points — so any team can score in every era they attempt, not just the ones that match their skill level exactly.

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02 — Suggested Question Categories (Subcommittee Picks)

Each Core member browses their era's pool below and, with their Subcom mentee, selects **2–4 questions** to actually build — mixing at least one Lite-friendly option with one harder option per era. These are starting points, not requirements; a subcommittee pair can propose a different topic in the same category if it fits the era better.

Era 0 — Warm-Up *(no tools beyond a browser/phone)* Cyber trivia · Spot the Phish · Terminology matching · Find-it-in-plain-sight (View Source) · Simple cipher decode (Caesar/ROT13/Atbash) · QR trail · Password

sense-check

Era I — Roots OSINT (profile digging, metadata, dorking, WHOIS/DNS) · Steganography (image LSB, audio spectrogram, whitespace) · Basic web (hidden comments, robots.txt, cookie decoding, single-step LFI) · Basic cryptography (Caesar, ROT13, stacked encoding, Morse) · Basic forensics (`strings` , magic-byte mismatch, simple `.pcap` extraction)

Era II — Settlement Auth bypass (exposed credentials, predictable sessions, cookie tampering) · SQL injection (login bypass, UNION-based extraction) · IDOR / enumeration (sequential IDs, undocumented endpoints) · Insecure file upload · Directory brute-forcing

Era III — Empire JWT forgery (weak secret, `alg: none`) · Privilege escalation (trusted client-side role parameters) · SSRF basics · CORS misconfiguration · Request smuggling · Exposed misconfig (`.git` , admin panels) · API abuse (rate-limit/business-rule bypass)

Era IV — Industry & Information Hash cracking (weak algorithms) · Stored XSS with a CSP gap · Insecure deserialization · Business-logic flaws (price/quantity manipulation) · Applied crypto flaws (padding oracle, weak keys) · Simple reverse-engineering "crackme"

Era V — Convergence One (or two, if bandwidth allows) full-chain scenario combining a foothold from an earlier era with a pivot into a second, internal-only service.

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03 — Points System

Team score = Σ(question points) − Σ(hint costs) + Σ(first-blood bonuses)

Question points, by era

Era	Point range per question
Warm-Up	50–75
Roots	75–100
Settlement	75–175 (Lite 75–100 · harder 150–175)
Empire	125–300 (Lite 125–150 · harder 250–300)
Industry & Information	300
Convergence	500

Hint costs (flat per era — applies to any question in that era)

Era	Hint cost
Warm-Up / Roots	10 pts
Settlement	15 pts
Empire	20 pts

Empire	20 pts
Industry & Information	25 pts
Convergence	No hints — mentors assist informally instead

One hint maximum per question.

First-blood bonus (per era, awarded once per question to the first correct solver)

Era	Bonus
Warm-Up / Roots	+5 pts
Settlement	+10 pts
Empire	+15 pts
Industry & Information	+20 pts
Convergence	+25 pts

Bonus is halved if the claiming team used a hint on that same question.

Tie-break: earliest timestamp of the deciding flag.

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04 — Awards

Run independently of the main leaderboard — a team can win one of these without placing top 3 overall.

Award	Basis
1st / 2nd / 3rd Overall	Main leaderboard score
Best Fresher Team	Highest score among all-fresher teams
The Pioneer	First team to fully clear an entire era
The Architect	Solved questions across the most different categories (breadth)
Speed Run	First team to reach 3 solved questions
Comeback of the Era	Biggest rank jump between the mid-event checkpoint and the final freeze
Night Owl	Correct flag closest to the final 10-minute mark
Cleanest Submission Log	Fewest incorrect attempts relative to questions solved
The Digger Award	Most questions attempted, solved or not
First Blood Champion	Most first-blood bonuses collected

The Convergence Award	Reached/solved anything in Era V (optional, gated era)
Most Creative Solve	Volunteer/judge vote
The Mentor's Pick	Each Core member nominates one standout team from their era
Spirit of Civilization	Volunteer vote for collaboration, fair play, energy

A team is eligible for multiple awards at once. All are announced in the same closing-ceremony slot.

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05 — Team Structure: Core & Subcom Mentorship

Each Core member owns one era (or category group), picks final questions from Section 02 with their Subcom mentee, and mentors that mentee through building and testing.

Core Member	Owns	Subcom Mentee	Together they
Core – Warm-Up/ Awareness	Era 0: trivia, phishing spotting, terminology, cipher basics	1	Pick 3–4 zero-barrier topics; subcom builds and user-tests on a non-technical friend
Core – OSINT/ Recon	Era I: OSINT, Basic Web	1	Pick 2–4 topics, subcom builds under review
Core – Crypto/ Forensics	Era I: Stego, Crypto, Forensics	1	Same pattern
Core – Web Security (Settlement)	Era II	1	Decide which get a Lite version; subcom builds both tiers
Core – Web Security (Empire)	Era III	1	Same pattern
Core – Applied Security	Era IV	1	Same pattern
Core – Infrastructure Lead	Deployment, CTFd config, Era V assembly	1 (infra-focused)	Subcom shadows the full server/Docker/ CTFd setup
Core – Dashboard & First-Blood Liaison	Live dashboard + first-blood notification (Section 06)	1	Acts as the committee's point of contact with the Web Development department; subcom helps test both features against staging
Core – Storyline	Flavor text, era intros, opening narration (Section 07)	1	Subcom drafts descriptions per era, Core edits for tone and theme consistency

	(Section 07)		
Core – Logistics & Platform	Points/hint tuning, CTFd entry, final testing (Section 07)	1	Subcom helps run the end-to-end test pass before go-live

A Core member reviewing their mentee's build before it goes to Infrastructure catches weak flags, broken hints, or mistimed difficulty earlier than a single collective dry run would.

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06 — Technical Requirements

Requirement	Description	Owner
CTFd server	Categories, questions, hints, team mode, flag checking	Infrastructure
Docker host	Runs all question containers	Infrastructure
Live dashboard	Real-time projector display of rankings and per-era progress, pulled from CTFd's API	**Web Development department**
First-blood notification	Automatic on-screen/audio alert the moment a question's first correct flag lands	**Web Development department**
Local network / hotspot	Connects participant laptops to the host	Infrastructure
Staging environment	Separate CTFd + challenge copy for pre-event testing only	Infrastructure
Backup host	Mirrored second laptop/server in case the primary fails	Infrastructure

Live dashboard brief: poll CTFd's `/api/v1/scoreboard` and `/api/v1/submissions` endpoints with an admin token; auto-refresh every 5–10s (no websockets needed); match the event's dark/era-themed visual style.

First-blood notification brief: CTFd has no native first-blood feature. Poll the submissions API, detect each question's first correct solve, and trigger a banner/sound on the dashboard. Test against the staging environment before event day.

CTFd notes: case-insensitive static flags; hints configured at the flat per-era cost above; Era V kept hidden and manually unlocked at the mid-event checkpoint; scoreboard freeze set to the event's end time.

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07 — Additional Roles

Storyline Writer — writes the flavor text for every question (tying it to its era's theme), writes each era's intro line and the opening-briefing narration, and keeps tone consistent across the whole set. Works from

intro line and the opening-briefing narration, and keeps tone consistent across the whole set. Works from the technical scenario each Core/Subcom pair supplies; doesn't alter the technical content itself.

Logistics & Platform Lead — owns the things that must stay consistent across every era rather than built per-category: finalizing point values and hint costs so the overall curve stays coherent, writing consistent hint wording, actually entering finished questions into CTFd (categories, flags, visibility), working with Infrastructure on server setup, and running the final end-to-end test pass before go-live.

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08 — Timeline

Pre-event

When	Task
T-3 weeks	Core/Subcom pairs pick final questions from Section 02; roles assigned
T-2 weeks	Questions built and delivered to Infrastructure for containerizing
T-3 days	Full dry run on staging: containers, CTFd, dashboard, first-blood, network
T-1 day	Final flag verification, print materials, close registration

Event day (2-hour run)

Time	Activity
T-20 to 0	Setup, registration, theme + inclusive briefing
00:00	Live. Era 0–II unlocked
00:30	Era III unlocks for all teams
00:50	Mid-event scoreboard reveal; Era V unlocks for top teams
01:40	Final warning, last hint round
01:50	Freeze, live flag reveal
02:00–02:10	Awards, closing, teardown

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09 — Classroom Items List

Item	Qty	Purpose
Participant laptops	1 per team min.	Challenge-solving
Host laptop + backup	2	Runs containers, CTFd, dashboard
Projector	1	Scoreboard, dashboard, timeline

Wi-Fi router / hotspot	1	Local network
Extension boards	3–4	Charging
Printed Quick-Start sheets	1 per team	Era 0/I onboarding
Sealed hint cards	3 per team	Physical hint ritual
Era banners	5–6	Visual theme
Name badges / team cards	All participants	Identification
Snacks + water	Per head	Energy over 2 hours
Certificates + prizes	All / top teams	Recognition

Estimated budget: **₹3,000–5,000**, mostly print materials and snacks — no dedicated server room or paid cloud VM required at this scale.